HW05 Tasks 1-2 - Performance Test Design, Execution, and AI Analysis Review

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Execution date: 2026-08-31 (Asia/Ho_Chi_Minh)
SUT: EShop backend from `../hw04/eshop-sut-main`
Selected flow: FR03 - FR09 - FR16
Tool: Apache JMeter 5.6.3, non-GUI execution

1. Scope and integrity statement

This report covers Tasks 1 and 2: source-grounded design, data-driven JMeter plans, smoke validation, real Load/Stress/Spike execution, a ten-minute soak test, raw evidence, a retained first-pass AI analysis, metric-misinterpretation review, and source-grounded optimization judgment. The selected flow is treated as the student's choice from the prompt; no independent group-allocation artifact was available, so this report does not falsely claim that uniqueness was externally verified.

All numerical results below were calculated from the full raw JTL files. Two contaminated/invalid attempts were preserved under `evidence/inconclusive` and excluded. No screenshot, voice recording, YouTube URL, GitHub Issue, or human-approval statement was generated by AI. The student-owned evidence boundary is stated in `STUDENT_EVIDENCE_REQUIRED.md`.

2. Test environment

Component	Actual value
Hostname	`MacBook-Air-cua-KunDa.local`
Hardware	MacBook Air `Mac16,12`, Apple M4, 10 cores, 16 GB RAM, arm64
Operating system	macOS 26.5.1 (25F80), Darwin 25.5.0
Backend	Node.js v26.4.0, Express 5.2.1, SQLite 3.51.0
JMeter	5.6.3, non-GUI mode
Main endpoint	`http://127.0.0.1:3000`
Isolated soak endpoint	`http://127.0.0.1:3001`, temporary copy of unchanged HW04 backend source and a separate SQLite file

The safe hardware table is in `evidence/hardware/hardware_spec.md`. Serial number, hardware UUID, and unrelated personal identifiers are not committed.

3. End-to-end workflow

Every Load, Stress, Spike, and Soak virtual user repeats the same six HTTP requests and the same assertions:

Order	Feature/group	Request	Correlation and business check
1	FR03 / auth-heavy	`POST /api/forgot-password`	HTTP 200; extract `$.resetToken`; require four digits.
2	FR03 / auth-heavy	`POST /api/reset-password`	Use the dynamic reset token; require exact success message.

Order	Feature/group	Request	Correlation and business check
3	FR03 / auth-heavy	`POST /api/login`	Extract JWT, user ID, and role; require non-empty JWT/user ID and `role=admin`.
4	FR09 / business validation	`POST /api/apply-coupon`	Correlate user ID; require `success=true`, discount 50,000 and final amount 550,000.
5	FR16 / transactional write	`POST /api/admin/import-products`	Send `Bearer ${jwt}`; require one insert, empty errors, and `1/1` message.
6	Read-heavy verification	`GET /api/products?search=...`	URL-encoded exact unique name; require the just-imported product in the JSON array.

FR16 is the transactional/write-heavy representative at the business-operation level. The source does not wrap a multi-row import in an SQLite transaction, so this report does not call it atomic or ACID. The final search supplies read-heavy coverage and validates the business effect of FR16.

4. Data, correlation, and assertions

`data/performance_users.csv` contains 80 synthetic accounts, exceeding the largest concurrency of 30 VUs. Each row supplies email/password, fixed `BIGBUY` coupon data, expected monetary values, and product fields. JWT, reset token, user ID, role, and the runtime-unique product name are never hard-coded in the CSV.

CSV sharing is `All threads`, recycling is explicit, and the dataset is larger than peak concurrency. A row may be reused in a later completed iteration, but the measured iteration timing and 80-row pool prevent concurrent reuse at 30 VUs; the zero correlation/business failures across the accepted runs provide an additional execution check. Eighty admin-role accounts are provisioned only after the canonical seed state is stable and are verified before JMeter begins.

Every sampler has an HTTP 200 assertion plus a JSON/business assertion. The fixed `BIGBUY` path avoids the source's known percentage-calculation defect, and `600000 > 500000` avoids the source's strict minimum-order boundary. View Results Tree is present only in the Spike plan and disabled during the high-load non-GUI measurement; the completed JTL can be loaded afterward.

5. Workload models

Plan	Model	Think time	Listener/report view	Rationale
`23127035_Load_20260831.jmx`	6 VUs, 15 s ramp, 120 s	Uniform 200-500 ms before each request	Summary Report	Reproduces a small steady interactive workload after smoke.
`23127035_Stress_20260831.jmx`	6, 12, then 24 VUs; 10 s ramp and 60 s per stage	Same	Aggregate Report	Doubles concurrency by stage and makes degradation attributable to a level.
`23127035_Spike_20260831.jmx`	3 VUs/45 s, 30 VUs/30 s with 1 s ramp, 3 VUs/45 s recovery	Same	View Results Tree, disabled during run	Creates an abrupt tenfold burst and observes recovery.
`23127035_Soak_20260831.jmx`	30 VUs, 30 s ramp, 600 s	Same	HTML dashboard/raw JTL	Uses the highest concurrency already proven error-free by the spike.

Stop/invalidity conditions were backend exit, JMeter non-zero exit, failed business assertions, non-zero error rate, missing raw/HTML output, or database state that could not be attributed to the plan. A measurement meeting none of those conditions is still a baseline, not an automatically approved SLA.

6. Execution and state control

The database file from HW04 was copied before any run with SHA-256 `86f8ab62d1f19c26f41fd62fd678bc8222eb383c189c5bac62d1b1de8b4564f4`. Before each accepted scenario the backend initialized its canonical 2 users, 5 products, and 4 coupons; the runner required that state to remain stable for three checks, inserted 80 performance accounts, verified exactly 80 admin accounts, and captured pre-state. The backend PID was sampled once per second for CPU and RSS.

The first Load attempt is excluded: the HTTP port opened before the destructive asynchronous initializer settled, which later removed the 80 accounts. The first Soak attempt is also excluded: deterministic post-counts proved unrelated traffic reached shared port 3000. Both full evidence sets remain under `evidence/inconclusive`. The accepted soak rerun used port 3001 and a separate temporary database to prevent local-client contamination.

No accepted run produced a failed login, so the SUT's lockout branch was not triggered. The reset procedure between runs was process stop, canonical SUT reinitialization, stability verification, account provisioning, and count verification; no silent in-run database edit occurred.

7. Accepted results

Percentiles use nearest rank over JMeter `elapsed` milliseconds. `Samples/s` counts HTTP samplers; one complete journey contains six samples, so `Workflows/s` is the approximate sample rate divided by six and is reported separately.

Scenario	Samples	Errors	Error %	Mean ms	p90 ms	p95 ms	p99 ms	Samples/s	Workflows/s	Peak CPU %	Peak RSS MB
Load	1,896	0	0.00	2.52	4	5	6	15.88	2.65	10.2	79.59
Stress	6,476	0	0.00	2.54	4	5	6	36.11	6.02	19.4	132.59
Spike	3,180	0	0.00	2.63	5	5	7	26.69	4.45	20.5	104.02
Soak	49,322	0	0.00	2.46	4	5	6	82.29	13.72	38.7	140.47

Stage-level evidence

Stage	Samples	Errors	p95 ms	Samples/s
Stress 6 VUs	936	0	5	15.78
Stress 12 VUs	1,854	0	5	31.10
Stress 24 VUs	3,686	0	5	61.87
Spike baseline 3 VUs	357	0	6	8.11
Spike burst 30 VUs	2,456	0	5	82.96
Spike recovery 3 VUs	367	0	5	8.21

Stress throughput scaled with concurrency while p95 remained 5 ms and the error rate remained zero. During Spike, recovery returned to 8.21 samples/s with p95 5 ms, comparable to the 8.11 samples/s/p95 6 ms baseline; no failure or persistent latency elevation was observed.

8. Database and resource consistency

Scenario	Products before	Successful FR16 samples	Products after	Coupon usage before/after	Consistent?
Load	5	315	320	0 / 0	Yes

Scenario	Products before	Successful FR16 samples	Products after	Coupon usage before/after	Consistent?
Stress	5	1,066	1,071	0 / 0	Yes
Spike	5	519	524	0 / 0	Yes
Soak isolated rerun	5	8,213	8,218	0 / 0	Yes

Resource figures are for the exact Node backend PID stored with each run, not whole-machine utilization. CPU percentages are macOS process percentages and are not normalized to the ten-core machine. RSS is resident memory, while the much larger VSZ field is not treated as physical memory usage.

9. Endurance conclusion

At 30 VUs for a full ten-minute measurement, the highest stable load observed was **82.29 HTTP samples/s**, approximately **13.72 complete workflows/s**, with p95 **5 ms**, p99 **6 ms**, **0.00% errors**, peak backend CPU **38.7%**, and peak RSS **140.47 MB**. The maximum individual elapsed time was **222 ms**, but it is an outlier and is not substituted for p95/p99.

This is a lower-bound empirical capacity statement for the tested envelope, not the absolute hardware breaking point: neither the 24-VU stress stage nor the 30-VU spike/soak caused failure or saturation. A higher stepped stress run would be needed to identify collapse; calling 30 VUs the absolute maximum would overstate the evidence.

10. AI review and corrections

Problem found	Correction applied	Evidence/reason
Shared CSV/header/timer initially existed only under thread group 1.	Moved them to Test Plan scope and re-ran smoke.	Later Stress/Spike groups otherwise had missing variables/timers.
Port readiness was mistaken for completed DB initialization.	Added stable canonical-count gate and exact 80-account verification.	Inconclusive Load attempt fell from 82 users to 2 during traffic.
Six-VU soak was arbitrary.	Raised to 30 VUs after the real spike passed.	Soak load now has an empirical prerequisite.
Shared port allowed unrelated local traffic during soak.	Excluded the attempt and reran on isolated port/database.	Product delta and coupon-usage rows could not be produced by the JTL.
JMeter summary throughput could be presented as user throughput.	Reported both HTTP samples/s and approximate six-request workflows/s.	Prevents a sixfold semantic overstatement.

These are AI self-audit corrections. They become the assignment's required **human review** only after the student checks the final artifacts and explicitly approves them; this report does not fabricate that approval.

11. Issue reporting

The accepted performance runs returned only HTTP 200 and had zero failed assertions; no crash, recovery failure, or measured performance regression was confirmed, so no new performance GitHub Issue was filed. Source limitations such as missing admin-role enforcement, the percentage-coupon formula, direct SQL interpolation, and destructive initialization are documented design observations, not falsely reported as new performance-run discoveries.

12. Deliverables and evidence map

- Test plans: `test-plans/23127035_{Load,Stress,Spike}_20260831.jmx`.
- Data: `data/performance_users.csv` and deterministic seed SQL.
- Raw results: `results/23127035_{Load,Stress,Spike}_20260831.jtl` plus the accepted Soak JTL.

- HTML dashboards: matching folders under `reports/`.
- Resource/PID/timestamps: `evidence/resource/`.
- Database pre/post snapshots: `evidence/database/`.
- Reproducible metrics: `analysis/task1_metrics.json` and `.md`.
- Hardware table: `evidence/hardware/hardware_spec.md`.
- Historical invalid runs: `evidence/inconclusive/` with reason and correction.
- AI process: `AI_Audit.md`, `AI_Critique.md`, and historical stage prompts/output under `audit/`.
- Reusable skill: `skills/jmeter-performance-testing/`.
- Student recording guide: `demo_video_script.md`.

Activity Monitor screenshots, the hardware screenshot, unlisted YouTube link with the student's Vietnamese voice, group-uniqueness evidence, and final human-review approval are not present because the assignment explicitly forbids AI fabrication of them. Exact handoff instructions are in `STUDENT_EVIDENCE_REQUIRED.md`.

13. Conclusion

The final Load, Stress, Spike, and isolated ten-minute Soak runs exercised one consistent, correlated FR03-FR09-FR16 workflow with distinct workload shapes and report views. All accepted samples passed HTTP and business assertions. Within the tested range, throughput scaled to 30 VUs without p95 degradation, errors, backend crash, or failed recovery. The conclusion is deliberately bounded to the measured machine, source revision, database state, think time, and date; it is not a universal SLA.

14. Task 2 AI output and review method

The unreviewed first-pass response is retained verbatim in `audit/task2_first_pass_ai_analysis.md`; it was not silently replaced with the corrected answer. `scripts/analyze_task2.py` independently parses the accepted JTLs, treats JMeter `success=false` as failure, uses nearest-rank percentiles over `elapsed`, and counts a completed workflow only when the successful final verification sampler exists. The complete audit is in `task2_analysis_review.md` and the reproducible values are in `analysis/task2_metrics.{md,json}`.

15. Task 2 corrected metrics

Scenario	Samples/s	Completed workflows	Completed workflows/s	p95 ms	p99 ms	Failed
Load	15.88	315	2.64	5	6	0
Stress	36.11	1,062	5.92	5	6	0
Spike	26.69	513	4.30	5	7	0
Soak	82.29	8,207	13.69	5	6	0

The Task 1 approximation `samples/s / 6` was useful for orientation but was not an exact completion measure because ramp-up/shutdown can leave partial iterations. Task 2 replaces it with the successful final-sampler count. The most material first-pass errors were calling 82.29 samples/s “workflows/s,” treating Stress/Spike whole-run averages as comparable peak stages, substituting the 222 ms maximum for p95/p99, and calling peak RSS a memory ceiling. Exact corrections and raw sources appear in the standalone Task 2 review.

16. Proposed regression gates, not SLA

On the same host/workflow/data/timer configuration, the proposed latency warning/fail bands are p95 above 7.5/10 ms and p99 above 10.5/14 ms. Throughput fail floors are 14.29 samples/s for Load 6 VUs, 55.68 for Stress 24 VUs, 74.66 for the Spike 30-VU burst, and 74.06 samples/s or 12.32 completed workflows/s for the 30-VU Soak. Any business assertion failure blocks the smoke gate. Node CPU above 50% or RSS above 170 MB triggers investigation only; these are host-dependent observations.

The bands represent proposed regression sensitivity (+50%/+100% latency and -10% throughput), not user requirements. Repeated clean runs must quantify natural variance before automation, and only the student/product owner can approve a formal threshold.

17. Optimization judgment

Source review supports experimenting with one explicit transaction around FR16 inserts, WAL plus `busy_timeout`, and a profiled non-unique `users(email)` index. It rejects generic advice to add a coupon-code index because `coupons.code` is already `UNIQUE`; rejects a normal B-tree name index as a direct fix for `LIKE '%term%'`; and rejects a client/server-style connection pool as a drop-in fix for the single SQLite handle. Search caching and clustered Node workers are also unsupported by the observed unique-write/read-after-write workload and may introduce stale reads or additional SQLite contention.

18. Task 2 conclusion and review ownership

The evidence establishes stable performance inside the tested envelope, not production readiness or absolute capacity. The Task 2 analysis/correction package is technically complete, reproducible, and keeps the original AI mistakes visible. The assignment's final human-review act remains the student's: the student must read the raw-log citations and explicitly approve or amend them. This report does not fabricate that approval.